
Parte 2: Procesamiento de consultas

1. Ejecución sin índices

1.1. Resultado Consulta 1

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {
      Edad: {
        '$gt': 50
      }
    },
    indexFilterSet: false,
    queryHash: '07B623A2',
    planCacheShapeHash: '07B623A2',
    planCacheKey: 'A139BEAA',
    optimizationTimeMillis: 47,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'PROJECTION_SIMPLE',
      transformBy: {
        _id: 0,
        Nombre: 1,
        Edad: 1,
        Nacionalidad: 1
      },
      inputStage: {
        stage: 'SORT',
        sortPattern: {
          Edad: 1
        },
        memLimit: 104857600,
        limitAmount: 10,
        type: 'simple',
        inputStage: {
          stage: 'COLLSCAN',
          filter: {
            Edad: {
              '$gt': 50
            }
          },
          direction: 'forward'
        }
      },
      rejectedPlans: []
    },
    executionStats: {
      executionSuccess: true,
```

```
nReturned: 7,
executionTimeMillis: 64,
totalKeysExamined: 0,
totalDocsExamined: 25,
executionStages: {
  isCached: false,
  stage: 'PROJECTION_SIMPLE',
  nReturned: 7,
  executionTimeMillisEstimate: 10,
  works: 34,
  advanced: 7,
  needTime: 26,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  transformBy: {
    _id: 0,
    Nombre: 1,
    Edad: 1,
    Nacionalidad: 1
  },
  inputStage: {
    stage: 'SORT',
    nReturned: 7,
    executionTimeMillisEstimate: 10,
    works: 34,
    advanced: 7,
    needTime: 26,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    sortPattern: {
      Edad: 1
    },
    memLimit: 104857600,
    limitAmount: 10,
    type: 'simple',
    totalDataSizeSorted: 4020,
    usedDisk: false,
    spills: 0,
    spilledDataStorageSize: 0,
    inputStage: {
      stage: 'COLLSCAN',
      filter: {
        Edad: {
          '$gt': 50
        }
      },
      nReturned: 7,
      executionTimeMillisEstimate: 10,
      works: 26,
      advanced: 7,
      needTime: 18,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      direction: 'forward',
      docsExamined: 25
    }
  }
}
```

```

    }
  },
  queryShapeHash: '21893A6C9EF92D4FCE3655C3939BC0D9115EAB01062C13D8FD87776F9FDFF6A3 ',
  command: {
    find: 'Voice-actors',
    filter: {
      Edad: {
        '$gt': 50
      }
    },
    sort: {
      Edad: 1
    },
    projection: {
      _id: 0,
      Nombre: 1,
      Edad: 1,
      Nacionalidad: 1
    },
    limit: 10,
    '$db': 'Actors'
  },
  serverInfo: {
    host: 'cecea55db233',
    port: 30002,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1788121883, i: 63 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1788121883, i: 63 })
}

```

1.2. Resultado Consulta 2

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {
      '$and': [
        {
          Idiomias: {
            '$eq': 'Ingles'
          }
        },
        {
          Nacionalidad: {
            '$eq': 'Chile'
          }
        }
      ]
    },
    indexFilterSet: false,
    queryHash: 'F3BD70B4',
    planCacheShapeHash: 'F3BD70B4',
    planCacheKey: 'BB3E0D7C',
    optimizationTimeMillis: 41,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'PROJECTION_SIMPLE',
      transformBy: {
        _id: 0,
        Nombre: 1,
        Nacionalidad: 1,
        Idiomias: 1
      },
      inputStage: {
        stage: 'SORT',
        sortPattern: {
          Nombre: 1
        },
        memLimit: 104857600,
        limitAmount: 10,
        type: 'simple',
        inputStage: {
          stage: 'COLLSCAN',
          filter: {
            '$and': [
              {
                Idiomias: {
                  '$eq': 'Ingles'
                }
              },
              {
                Nacionalidad: {
                  '$eq': 'Chile'
                }
              }
            ]
          },
          direction: 'forward'
        }
      }
    }
  }
}
```

```

    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 6,
  executionTimeMillis: 60,
  totalKeysExamined: 0,
  totalDocsExamined: 25,
  executionStages: {
    isCached: false,
    stage: 'PROJECTION_SIMPLE',
    nReturned: 6,
    executionTimeMillisEstimate: 10,
    works: 33,
    advanced: 6,
    needTime: 26,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    transformBy: {
      _id: 0,
      Nombre: 1,
      Nacionalidad: 1,
      Idiomas: 1
    },
    inputStage: {
      stage: 'SORT',
      nReturned: 6,
      executionTimeMillisEstimate: 10,
      works: 33,
      advanced: 6,
      needTime: 26,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      sortPattern: {
        Nombre: 1
      },
      memLimit: 104857600,
      limitAmount: 10,
      type: 'simple',
      totalDataSizeSorted: 3702,
      usedDisk: false,
      spills: 0,
      spilledDataStorageSize: 0,
      inputStage: {
        stage: 'COLLSCAN',
        filter: {
          '$and': [
            {
              Idiomas: {
                '$eq': 'Ingles'
              }
            },
            {
              Nacionalidad: {
                '$eq': 'Chile'
              }
            }
          ]
        }
      }
    }
  }
}

```

```

        }
    ]
},
nReturned: 6,
executionTimeMillisEstimate: 0,
works: 26,
advanced: 6,
needTime: 19,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
direction: 'forward',
docsExamined: 25
    }
}
},
queryShapeHash: 'FBD5590A7B8E5C87163EAA81455789FD2C4973D8175382668E2C98FBFD3C289F ',
command: {
    find: 'Voice-actors',
    filter: {
        Nacionalidad: 'Chile',
        Idiomas: 'Ingles'
    },
    sort: {
        Nombre: 1
    },
    projection: {
        _id: 0,
        Nombre: 1,
        Nacionalidad: 1,
        Idiomas: 1
    },
    limit: 10,
    '$db': 'Actors'
},
serverInfo: {
    host: 'cecea55db233',
    port: 30002,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
    clusterTime: Timestamp({ t: 1788123725, i: 1 }),
    signature: {
        hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
        keyId: Long('0')
    }
}
},

```

```
operationTime: Timestamp({ t: 1788123725, i: 1 })
}
```

1.3. Resultado Consulta 3

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {
      '$and': [
        {
          Personajes: {
            '$elemMatch': {
              '$and': [
                {
                  Generos: {
                    '$eq': 'Aventura'
                  }
                },
                {
                  Rol: {
                    '$eq': 'principal'
                  }
                }
              ]
            }
          },
          Edad: {
            '$lt': 30
          }
        ]
      ],
      indexFilterSet: false,
      queryHash: 'E4BD0639',
      planCacheShapeHash: 'E4BD0639',
      planCacheKey: '29D7E03D',
      optimizationTimeMillis: 24,
      maxIndexedOrSolutionsReached: false,
      maxIndexedAndSolutionsReached: false,
      maxScansToExplodeReached: false,
      prunedSimilarIndexes: false,
      winningPlan: {
        isCached: false,
        stage: 'PROJECTION_DEFAULT',
        transformBy: {
          _id: 0,
          Nombre: 1,
          Edad: 1,
          'Personajes.Rol': 1,
          'Personajes.Generos': 1
        }
      },
      inputStage: {
        stage: 'SORT',
        sortPattern: {
          Edad: 1
        }
      },

```

```

memLimit: 104857600,
limitAmount: 10,
type: 'simple',
inputStage: {
  stage: 'COLLSCAN',
  filter: {
    '$and': [
      {
        Personajes: {
          '$elemMatch': {
            '$and': [
              {
                Generos: {
                  '$eq': 'Aventura'
                }
              },
              {
                Rol: {
                  '$eq': 'principal'
                }
              }
            ]
          }
        },
        {
          Edad: {
            '$lt': 30
          }
        }
      ]
    },
    direction: 'forward'
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 4,
  executionTimeMillis: 32,
  totalKeysExamined: 0,
  totalDocsExamined: 25,
  executionStages: {
    isCached: false,
    stage: 'PROJECTION_DEFAULT',
    nReturned: 4,
    executionTimeMillisEstimate: 0,
    works: 31,
    advanced: 4,
    needTime: 26,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    transformBy: {
      _id: 0,
      Nombre: 1,
      Edad: 1,
      'Personajes.Rol': 1,
      'Personajes.Generos': 1
    }
  }
},

```

```

inputStage: {
  stage: 'SORT',
  nReturned: 4,
  executionTimeMillisEstimate: 0,
  works: 31,
  advanced: 4,
  needTime: 26,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  sortPattern: {
    Edad: 1
  },
  memLimit: 104857600,
  limitAmount: 10,
  type: 'simple',
  totalDataSizeSorted: 2345,
  usedDisk: false,
  spills: 0,
  spilledDataStorageSize: 0,
  inputStage: {
    stage: 'COLLSCAN',
    filter: {
      '$and': [
        {
          Personajes: {
            '$elemMatch': {
              '$and': [
                {
                  Generos: {
                    '$eq': 'Aventura'
                  }
                },
                {
                  Rol: {
                    '$eq': 'principal'
                  }
                }
              ]
            }
          }
        }
      ],
      Edad: {
        '$lt': 30
      }
    }
  },
  nReturned: 4,
  executionTimeMillisEstimate: 0,
  works: 26,
  advanced: 4,
  needTime: 21,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  direction: 'forward',
  docsExamined: 25
}
}

```

```

    }
  },
  queryShapeHash: '74B9733C635319C8A04021D3B8412F204CDBBF8F31453A13449D532A3877E557 ',
  command: {
    find: 'Voice-actors',
    filter: {
      Edad: {
        '$lt': 30
      },
      Personajes: {
        '$elemMatch': {
          Rol: 'principal',
          Generos: 'Aventura'
        }
      }
    },
    sort: {
      Edad: 1
    },
    projection: {
      _id: 0,
      Nombre: 1,
      Edad: 1,
      'Personajes.Rol': 1,
      'Personajes.Generos': 1
    },
    limit: 10,
    '$db': 'Actors'
  },
  serverInfo: {
    host: 'cecea55db233',
    port: 30002,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1788124889, i: 7 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1788124889, i: 7 })
}

```

1.4. Resultado Consulta 4

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {},
    indexFilterSet: false,
    queryHash: 'B14C1969',
    planCacheShapeHash: 'B14C1969',
    planCacheKey: '5525452D',
    optimizationTimeMillis: 14,
    optimizedPipeline: true,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'PROJECTION_SIMPLE',
      transformBy: {
        Nombre: true,
        Edad: true,
        Patrimonio: true,
        _id: false
      },
      inputStage: {
        stage: 'SKIP',
        skipAmount: 4,
        inputStage: {
          stage: 'SORT',
          sortPattern: {
            Patrimonio: 1
          },
          memLimit: 104857600,
          limitAmount: 14,
          type: 'simple',
          inputStage: {
            stage: 'COLLSCAN',
            direction: 'forward'
          }
        }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 10,
    executionTimeMillis: 56,
    totalKeysExamined: 0,
    totalDocsExamined: 25,
    executionStages: {
      isCached: false,
      stage: 'PROJECTION_SIMPLE',
      nReturned: 10,
      executionTimeMillisEstimate: 9,
      works: 41,
      advanced: 10,
      needTime: 30,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
    }
  }
}
```

```

isEOF: 1,
transformBy: {
  Nombre: true,
  Edad: true,
  Patrimonio: true,
  _id: false
},
inputStage: {
  stage: 'SKIP',
  nReturned: 10,
  executionTimeMillisEstimate: 9,
  works: 41,
  advanced: 10,
  needTime: 30,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  skipAmount: 4,
  inputStage: {
    stage: 'SORT',
    nReturned: 14,
    executionTimeMillisEstimate: 9,
    works: 41,
    advanced: 14,
    needTime: 26,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    sortPattern: {
      Patrimonio: 1
    },
    memLimit: 104857600,
    limitAmount: 14,
    type: 'simple',
    totalDataSizeSorted: 8090,
    usedDisk: false,
    spills: 0,
    spilledDataStorageSize: 0,
    inputStage: {
      stage: 'COLLSCAN',
      nReturned: 25,
      executionTimeMillisEstimate: 0,
      works: 26,
      advanced: 25,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      direction: 'forward',
      docsExamined: 25
    }
  }
}
},
queryShapeHash: '1A15F0DBB602997BF9D986C82E3524428930E640CEF877D98EAA877D9AE620A8 ',
command: {
  aggregate: 'Voice-actors',
  pipeline: [
    {

```

```

    '$sort': {
      Patrimonio: 1
    }
  },
  {
    '$skip': 4
  },
  {
    '$limit': 10
  },
  {
    '$project': {
      _id: 0,
      Nombre: 1,
      Edad: 1,
      Patrimonio: 1
    }
  }
],
cursor: {},
'$db': 'Actors'
},
serverInfo: {
  host: 'cecea55db233',
  port: 30002,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1788125265, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
    keyId: Long('0')
  }
},
operationTime: Timestamp({ t: 1788125265, i: 1 })
}

```